**TASK: 1**

## Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

**Command:**

```
ping www.youtube.com
```

**Sample Output:**

```
Pinging youtube.com [142.250.183.14] with 32 bytes of
data:
Reply from 142.250.183.14: bytes=32 time=25ms TTL=117
Reply from 142.250.183.14: bytes=32 time=28ms TTL=117
Reply from 142.250.183.14: bytes=32 time=24ms TTL=117
Reply from 142.250.183.14: bytes=32 time=27ms TTL=117

Ping statistics for 142.250.183.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
Approximate round trip times in milli-seconds:
    Minimum = 24ms, Maximum = 28ms, Average = 26ms
```

**Result:** The ping test was successful, with 4 packets received and 0% packet loss. The sample average response time was **26 milliseconds (ms)**.

## TASK: 2

## Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.<br><br>Hint: Each hop represents a network device your data passes through to reach the destination.

**Command:**

```
tracert www.flipkart.com
```

**Sample Output:**

```
Tracing route to www.flipkart.com
over a maximum of 30 hops:

  1     2 ms     1 ms     2 ms  192.168.1.1
  2    10 ms    12 ms    11 ms  10.10.0.1
  3    18 ms    20 ms    19 ms  100.64.0.1
  4    25 ms    23 ms    26 ms  203.0.113.10
  5    30 ms    28 ms    31 ms  * * *
  6    35 ms    33 ms    34 ms  Destination reached
```

**Intermediate hops observed:**

| Hop | IP Address / Device | Description |
|---|---|---|
| 1 | 192.168.1.1 | Local Wi-Fi router |
| 2 | 10.10.0.1 | Example ISP network device |
| 3 | 100.64.0.1 | Example ISP routing device |
| 4 | 203.0.113.10 | Illustrative intermediate address |
| 5 | Request timed out | Router did not respond to the probe |
| 6 | Destination reached | Illustrative endpoint |

NETWORK

**Result:** The `tracert` command displays the intermediate network hops between my computer and the destination. Some routers may not respond, so a timeout does not necessarily mean the connection has failed.

## TASK: 3

## Use the nslookup tool to find the IP address of www.zomato.com and record the result.

**Command:**

```
nslookup www.zomato.com
```

**Sample Output:**

```
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    www.zomato.com
Addresses:  104.18.32.169
            172.64.155.87
```

**Result:** The `nslookup` command queries the DNS system to find the IP address associated with a domain name. In this example, the returned IP addresses are `104.18.32.169` and `172.64.155.87`.

NETWORK

## TASK: 4

## Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

**Command:**

```
ping 192.168.1.5
```

Here, `192.168.1.5` is an example IP address assigned to a friend's laptop or mobile device on the same local network.

**Sample Output when the device is online:**

```
Pinging 192.168.1.5 with 32 bytes of data:
Reply from 192.168.1.5: bytes=32 time=3ms TTL=128
Reply from 192.168.1.5: bytes=32 time=2ms TTL=128
Reply from 192.168.1.5: bytes=32 time=3ms TTL=128
Reply from 192.168.1.5: bytes=32 time=2ms TTL=128
```

**What happens when the device is offline?**

If the device is switched off or disconnected from the network, the ping command may display `Request timed out` or `Destination host unreachable`. This indicates that the device is not responding to the ping request. However, some devices block ping requests even when they are online.

**Result:** The ping command helps check connectivity between two devices on the same local network.

NETWORK

## TASK: 5

## Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.<br><br>Constraint: Give a specific example for each tool based on your own test results above.

1. **Ping:** The ping command is useful for checking whether a website or device responds over the network. In my YouTube test, an illustrative average response time of 26 ms indicated a quick response with no packet loss.
2. **Traceroute:** The tracert command helps identify the network path between a computer and a destination. In my Flipkart example, it showed intermediate hops and a timeout, which can help identify where responses stop.
3. **Nslookup:** The nslookup command is useful for diagnosing DNS problems by checking whether a domain name resolves to an IP address. In my Zomato example, it returned IP addresses associated with the domain.
4. **Website not loading:** I would use ping to check connectivity and tracert to investigate where the route may be experiencing problems.
5. **Slow connection:** I would use ping to check response times and packet loss, then tracert to identify hops with unusually high delays.
6. **Suspected DNS issue:** I would use nslookup to check whether the domain resolves correctly. If DNS resolution fails, I would investigate the DNS server and network configuration.